

OPTIMISATION OF WIDE, SHALLOW NEURAL NETWORKS BY GRADIENT FLOW: AN ELEMENTARY ANALYSIS

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ABSTRACT. Despite their non-convex loss landscape, sufficiently wide neural networks with one hidden layer can fit a finite data set by gradient flow. We give an elementary analysis of this result under the assumption of smooth activations and fixed output weights. Two differential inequalities relate the decay of the training residual to the variation of the empirical tangent kernel. These inequalities exhibit a quantity which is monotonically non-decreasing under the flow, whose initial positivity prevents the kernel from degenerating, and which gives explicit bounds on the residual, loss, and total kernel motion. The argument is deterministic and holds at sufficiently large finite width. An example with Gaussian design matrix explains the sizes of the constants and gives a sufficient width linear in the sample size when the population kernel is well conditioned, up to confidence factors.

1. INTRODUCTION

Consider fitting a two-layer neural network to a finite data set by gradient flow. For square loss, the vector of predictions evolves over time according to the empirical tangent kernel of the network. If the smallest eigenvalue of this kernel remains bounded uniformly away from zero, then the residual decays exponentially towards zero, and the predictions converge to the training responses. This observation is the starting point of many analyses in the neural tangent kernel regime.

The delicate point in these analyses is that the kernel itself changes during training. Its stability and residual decay seem to depend on one another: a stable kernel drives the residual down, while a small residual keeps the weights, and therefore the kernel, from moving too far.

We express this feedback through two scalar quantities. Writing r_t for the residual norm and λ_t for the minimum eigenvalue of the kernel matrix, the network dynamics imply differential inequalities of the form

$$\dot{r}_t \leq -\lambda_t r_t, \quad \dot{\lambda}_t \geq -\beta r_t,$$

for an explicit $\beta \geq 0$. Combining these inequalities shows that the quantity $\lambda_t^2 - 2\beta r_t$ is nondecreasing in time. As a consequence, any decrease in curvature must be accompanied by a corresponding decrease in the residual. In particular, if this quantity is initially positive, then the curvature remains uniformly positive for all times. A Riccati comparison sharpens this qualitative conclusion to explicit time-dependent bounds on r_t (from above) and λ_t (from below).

The first part of the note develops this deterministic argument for smooth activations and arbitrary fixed output weights. The second part asks how large its constants are under a basic probabilistic model. For normalised Gaussian covariates and order-one responses, the calculation reveals $\sqrt{n/m}$ as the pertinent ‘small’ parameter.

2. SETUP AND NOTATION

All vector norms are Euclidean, matrix norms carry the subscript op when they are operator norms, and gradients in the hidden weights use the Euclidean norm on all their entries.

Let $x_1, \dots, x_n \in \mathbb{R}^d$ be fixed training inputs, and let $y \in \mathbb{R}^n$ be the target vector. Write $X \in \mathbb{R}^{n \times d}$ for the matrix with rows $x_i^\top$ and $G = XX^\top \in \mathbb{R}^{n \times n}$ for the associated Gram matrix. Fix arbitrary output weights $a_1, \dots, a_m \in \mathbb{R}$ and consider the parameterised regression function

$$f_W(x) = \frac{1}{\sqrt{m}} \sum_{j=1}^m a_j \sigma(w_j^\top x), \quad W = (w_1, \dots, w_m).$$

In what follows, only the hidden weights W are trained. Assume that $\sigma \in C^2(\mathbb{R})$ and satisfies the uniform smoothness bounds

$$\sup_{z \in \mathbb{R}} |\sigma'(z)| \leq B_1, \quad \sup_{z \in \mathbb{R}} |\sigma''(z)| \leq B_2.$$

The prediction vector, residual vector, and loss are defined as

$$F(W) := \begin{pmatrix} f_W(x_1) \\ \vdots \\ f_W(x_n) \end{pmatrix}, \quad e(W) := F(W) - y, \quad \mathcal{L}(W) := \frac{1}{2} \|e(W)\|^2.$$

We study fitting this model in continuous time by the gradient flow dynamics

$$\dot{W}_t = -\nabla \mathcal{L}(W_t).$$

For each neuron j , define the diagonal matrix

$$D_j(W) := \text{diag}(\sigma'(x_1^\top w_j), \dots, \sigma'(x_n^\top w_j)) \in \mathbb{R}^{n \times n}.$$

Let $J(W) := D_W F(W)$ denote the Jacobian of the prediction vector. Its block associated with w_j is

$$J_j(W) = \frac{a_j}{\sqrt{m}} D_j(W) X \in \mathbb{R}^{n \times d}.$$

Collecting these blocks gives the empirical tangent kernel

$$K(W) = J(W)J(W)^\top = \frac{1}{m} \sum_{j=1}^m a_j^2 D_j(W) G D_j(W) \succeq 0.$$

Along the flow, set

$$e_t := e(W_t), \quad K_t := K(W_t), \quad r_t := \|e_t\|, \quad \lambda_t := \lambda_{\min}(K_t).$$

Thus r_t is the residual norm, while λ_t is the smallest eigenvalue of the tangent kernel; we refer to this as the minimum curvature.

3. THE SELF-BOUNDING ARGUMENT

The argument rests on two observations. First, the residual obeys a closed evolution equation once the current tangent kernel is known. Second, smoothness prevents that kernel from moving faster than a constant multiple of the residual norm. We establish these facts in turn before combining them in a scalar comparison.

Lemma 1. *Along gradient flow, the residual satisfies*

$$\dot{e}_t = -K_t e_t.$$

Consequently,

$$\frac{d}{dt} \frac{1}{2} \|e_t\|^2 = -e_t^\top K_t e_t \leq -\lambda_t \|e_t\|^2,$$

and, for almost every t ,

$$\dot{r}_t \leq -\lambda_t r_t.$$

Proof. The chain rule and $\nabla \mathcal{L}(W_t) = J(W_t)^\top e_t$ give

$$\dot{e}_t = J(W_t)\dot{W}_t = -J(W_t)J(W_t)^\top e_t = -K_t e_t.$$

Taking the inner product with e_t proves the energy identity. The inequality for r_t follows wherever $r_t > 0$; if the residual reaches zero, it remains zero. $\square$

In particular, the residual r_t is nonincreasing. Together with the bound on σ' , this prevents finite-time blow-up of the hidden weights, so the gradient flow solution exists for all time.

We next control the motion of the kernel K_t . The fixed output weights enter through their empirical third absolute moment

$$\nu_3(a) := \frac{1}{m} \sum_{j=1}^m |a_j|^3.$$

We also set

$$C_X := 2B_1^2 B_2 \|X\|_{\text{op}}^4, \quad \beta_{m,a} := \frac{C_X \nu_3(a)}{\sqrt{m}}.$$

Lemma 2. *Along gradient flow,*

$$\|\dot{K}_t\|_{\text{op}} \leq \beta_{m,a} r_t.$$

Moreover, the function λ_t is absolutely continuous and, for almost every t ,

$$\dot{\lambda}_t \geq -\beta_{m,a} r_t.$$

Proof. Write $D_{j,t} := D_j(W_t)$. The velocity of the j th hidden weight is

$$\dot{w}_{j,t} = -\frac{a_j}{\sqrt{m}} X^\top D_{j,t} e_t,$$

and therefore

$$\|\dot{w}_{j,t}\| \leq \frac{|a_j| B_1 \|X\|_{\text{op}}}{\sqrt{m}} r_t.$$

Since

$$\dot{D}_{j,t} = \text{diag}_{1 \leq i \leq n} (\sigma''(x_i^\top w_{j,t}) x_i^\top \dot{w}_{j,t}),$$

we also have

$$\|\dot{D}_{j,t}\|_{\text{op}} \leq B_2 \|X \dot{w}_{j,t}\| \leq \frac{|a_j| B_1 B_2 \|X\|_{\text{op}}^2}{\sqrt{m}} r_t.$$

Differentiating the kernel gives

$$\dot{K}_t = \frac{1}{m} \sum_{j=1}^m a_j^2 (\dot{D}_{j,t} G D_{j,t} + D_{j,t} G \dot{D}_{j,t}).$$

Using $\|D_{j,t}\|_{\text{op}} \leq B_1$ and $\|G\|_{\text{op}} = \|X\|_{\text{op}}^2$ gives

$$\|\dot{K}_t\|_{\text{op}} \leq \frac{2B_1^2 B_2 \|X\|_{\text{op}}^4}{m\sqrt{m}} \left(\sum_{j=1}^m |a_j|^3 \right) r_t = \beta_{m,a} r_t.$$

Finally, the smallest eigenvalue is a Lipschitz function of a symmetric matrix. It follows that λ_t is absolutely continuous and, for almost every t , satisfies $\dot{\lambda}_t \geq -\|\dot{K}_t\|_{\text{op}}$. $\square$

The preceding lemmas reduce the network dynamics to two scalar differential inequalities. The residual both drives kernel motion and decays under the influence of the kernel's curvature. The next proposition combines these effects, demonstrating that sufficiently large initial curvature limits the total residual, and therefore its own subsequent variation.

Proposition 3. *Let $r, \lambda : [0, \infty) \rightarrow [0, \infty)$ be absolutely continuous.*

(1) *Suppose there is a constant $\beta \geq 0$ such that, for almost every t ,*

$$\dot{r}_t \leq -\lambda_t r_t, \quad \dot{\lambda}_t \geq -\beta r_t.$$

Then

$$\lambda_t^2 - 2\beta r_t \geq \lambda_0^2 - 2\beta r_0$$

for every $t \geq 0$.

(2) *Suppose in addition that*

$$\kappa := \sqrt{\lambda_0^2 - 2\beta r_0} > 0, \quad \rho := \frac{\lambda_0 - \kappa}{\lambda_0 + \kappa} \in [0, 1).$$

Define

$$\bar{\lambda}_t := \kappa \frac{1 + \rho e^{-\kappa t}}{1 - \rho e^{-\kappa t}}, \quad \bar{r}_t := r_0 e^{-\kappa t} \left(\frac{1 - \rho}{1 - \rho e^{-\kappa t}} \right)^2.$$

Then, for every $t \geq 0$,

$$\lambda_t \geq \bar{\lambda}_t \geq \kappa, \quad r_t \leq \bar{r}_t \leq r_0 e^{-\kappa t}.$$

Furthermore,

$$\int_0^\infty r_t dt \leq \frac{2r_0}{\lambda_0 + \kappa},$$

and, for every $t \geq 0$,

$$\beta \int_0^t r_s ds \leq \lambda_0 - \bar{\lambda}_t \leq \lambda_0 - \kappa.$$

Proof. Differentiating the proposed monotone quantity gives, for almost every time,

$$\frac{d}{dt}(\lambda^2 - 2\beta r) = 2\lambda\dot{\lambda} - 2\beta\dot{r} \geq -2\beta\lambda r + 2\beta\lambda r = 0,$$

which proves the first assertion. If $\beta = 0$, then $\lambda_t \geq \lambda_0$ and $r_t \leq r_0 e^{-\lambda_0 t}$, so the remaining conclusions are immediate. We may therefore assume $\beta > 0$. Under the assumption that $\kappa > 0$, the monotone quantity gives the bound $\lambda_t \geq \sqrt{\kappa^2 + 2\beta r_t}$, and hence the nonlinear comparison

$$\dot{r}_t \leq -r_t \sqrt{\kappa^2 + 2\beta r_t}.$$

Let R solve the comparison equation $\dot{R} = -R\sqrt{\kappa^2 + 2\beta R}$, $R_0 = r_0$. Scalar comparison gives $r_t \leq R_t$. Writing $\Lambda = \sqrt{\kappa^2 + 2\beta R}$ (equivalently, $R = \frac{\Lambda^2 - \kappa^2}{2\beta}$), differentiation gives $\dot{\Lambda} = -\beta R = \frac{\kappa^2 - \Lambda^2}{2}$ and $\Lambda_0 = \lambda_0$. Its solution is exactly $\bar{\lambda}_t$. Substitution gives $R_t = \bar{r}_t$, and therefore $r_t \leq \bar{r}_t$.

Finally, since

$$\dot{\lambda} \geq -\beta r \geq -\beta \bar{r} = \dot{\bar{\lambda}}, \quad \lambda_0 = \bar{\lambda}_0,$$

integration gives $\lambda_t \geq \bar{\lambda}_t$, as well as

$$\beta \int_0^t r_s ds \leq \beta \int_0^t \bar{r}_s ds = \lambda_0 - \bar{\lambda}_t.$$

Letting $t \rightarrow \infty$ and using $\lambda_0^2 - \kappa^2 = 2\beta r_0$ gives

$$\int_0^\infty r_t dt \leq \frac{\lambda_0 - \kappa}{\beta} = \frac{2r_0}{\lambda_0 + \kappa}.$$

□

Remark 4. At the nontrivial borderline $\lambda_0^2 = 2\beta r_0$, with $\beta > 0$ and $\lambda_0 > 0$, the same comparison holds with

$$\bar{\lambda}_t = \frac{\lambda_0}{1 + \lambda_0 t/2}, \quad \bar{r}_t = \frac{r_0}{(1 + \lambda_0 t/2)^2}.$$

The comparison no longer supplies a strictly positive lower bound on the curvature that is uniform in time, as $\bar{\lambda}_t$ tends to zero for large t . The residual bound nevertheless forces r_t to vanish at the polynomial rate $O(t^{-2})$, and the loss at the rate $O(t^{-4})$.

Theorem 5. *Consider the two-layer network and gradient flow above. Let*

$$\lambda_0 := \lambda_{\min}(K(W_0)), \quad r_0 := \|F(W_0) - y\|, \quad \beta_{m,a} := \frac{2B_1^2 B_2 \|X\|_{\text{op}}^4 \nu_3(a)}{\sqrt{m}}.$$

Suppose that

$$\lambda_0^2 > 2\beta_{m,a} r_0,$$

and define κ , ρ , $\bar{\lambda}_t$, and $\bar{r}_t$ as in the preceding proposition with $\beta = \beta_{m,a}$. Then, for every $t \geq 0$,

$$\lambda_{\min}(K(W_t)) \geq \bar{\lambda}_t \geq \kappa > 0, \quad \|e_t\| = r_t \leq \bar{r}_t \leq r_0 e^{-\kappa t}.$$

The loss satisfies

$$\mathcal{L}(W_t) \leq \mathcal{L}(W_0) e^{-2\kappa t} \left(\frac{1 - \rho}{1 - \rho e^{-\kappa t}} \right)^4 \leq \mathcal{L}(W_0) e^{-2\kappa t}.$$

Moreover,

$$\int_0^\infty r_t dt \leq \frac{2r_0}{\lambda_0 + \kappa},$$

and the kernel motion satisfies the finite-time estimate

$$\|K(W_t) - K(W_0)\|_{\text{op}} \leq \lambda_0 - \bar{\lambda}_t \leq \lambda_0 - \kappa = \frac{2\beta_{m,a} r_0}{\lambda_0 + \kappa}.$$

Proof. The residual and kernel-variation lemmas verify the hypotheses of the scalar proposition with $r = r_t$, $\lambda = \lambda_t$, and $\beta = \beta_{m,a}$. They therefore give all asserted curvature, residual, and integral estimates. Squaring the full Riccati envelope for the residual gives the sharpened loss bound. Finally,

$$\|K_t - K_0\|_{\text{op}} \leq \int_0^t \|\dot{K}_s\|_{\text{op}} ds \leq \beta_{m,a} \int_0^t r_s ds,$$

and the last claim follows from the finite-time integral estimate in the proposition. □

4. TYPICAL CONSTANTS UNDER RANDOM INITIALISATION

The theorem is deterministic, but its condition is most useful when the quantities appearing in it can be estimated before training. We first control the initial empirical curvature conditional on the data, and then work through a simple probabilistic model for the covariates and responses.

4.1. Concentration of λ_0 . Suppose that $w_1(0), \dots, w_m(0)$ are independent and identically distributed, while the fixed output weights are deterministic or independent of the hidden initialisation. Define

$$D(w) := \text{diag}_{1 \leq i \leq n}(\sigma'(x_i^\top w)), \quad K_\infty := \mathbb{E}[D(w)GD(w)],$$

and suppose that $\lambda_\infty := \lambda_{\min}(K_\infty) > 0$. Also set

$$\nu_2(a) := \frac{1}{m} \sum_{j=1}^m a_j^2, \quad A_{\max} := \max_{1 \leq j \leq m} |a_j|,$$

and assume $\nu_2(a) > 0$. Conditional on the output weights,

$$\mathbb{E}[K(W_0)] = \nu_2(a)K_\infty.$$

Since $0 \preceq D(w)GD(w) \preceq B_1^2 \|X\|_{\text{op}}^2 I$ almost surely, the matrix Chernoff inequality shows that, for $0 < \delta, \eta < 1$, it is enough to take

$$m \gtrsim \frac{A_{\max}^2 B_1^2 \|X\|_{\text{op}}^2}{\delta^2 \nu_2(a) \lambda_\infty} \log \frac{n}{\eta},$$

to ensure, with probability at least $1 - \eta$,

$$\lambda_{\min}(K(W_0)) \geq (1 - \delta) \nu_2(a) \lambda_\infty.$$

That is, a sufficiently wide initialisation inherits, up to a factor $1 - \delta$, the curvature of its infinite-width limit.

4.2. Gaussian covariates. The remaining constants are easiest to interpret after fixing the scale of the data. Consider the normalised Gaussian design

$$x_i = d^{-1/2} z_i, \quad z_i \stackrel{\text{iid}}{\sim} N(0, I_d), \quad \gamma := \frac{n}{d} < 1.$$

For any $u > 0$ such that $\sqrt{\gamma} + u < 1$, the usual Gaussian singular-value bounds give, with probability at least $1 - 2e^{-du^2/2}$,

$$\|X\|_{\text{op}} \leq L_\gamma := 1 + \sqrt{\gamma} + u, \quad \lambda_{\min}(G) \geq \ell_\gamma^2, \quad \ell_\gamma := 1 - \sqrt{\gamma} - u.$$

On this event,

$$C_X \leq C_\gamma := 2B_1^2 B_2 L_\gamma^4.$$

This is the benefit of the $d^{-1/2}$ normalisation: for unnormalised Gaussian covariates, $\|X\|_{\text{op}}$ is of order $\sqrt{d} + \sqrt{n}$, and the displayed estimate for C_X carries the fourth power of that scale.

For a concrete example, take $\sigma = \tanh$, for which we may use $B_1 = 1$ and $B_2 = 2$. Initialise the hidden weights independently with $w_j(0) \sim N(0, I_d)$, independently of the data. Conditional on the data, the inequality

$$D(w)GD(w) \succeq \lambda_{\min}(G)D(w)^2$$

gives

$$K_\infty \succeq \lambda_{\min}(G) \mathbb{E}_w[D(w)^2].$$

Each diagonal entry on the right involves $\mathbb{E}_w[\text{sech}^4(w^\top x_i)]$. On the preceding data event, $\|x_i\| \leq L_\gamma$. Since $w^\top x_i$ has the same distribution as $\|x_i\|Z$ for $Z \sim N(0, 1)$, and $\text{sech}^4(z)$ decreases with $|z|$, we obtain

$$\lambda_\infty \geq \lambda_* := \ell_\gamma^2 \mathbb{E}[\text{sech}^4(L_\gamma Z)] > 0.$$

This controls the initial population curvature. Under the width condition below, the self-bounding argument then keeps the empirical curvature positive during training. The restriction $n < d$ is used only for this elementary lower bound through G ; a nonlinear population kernel can remain positive definite when $n \geq d$.

Now take independent symmetric Rademacher output weights, independent of the data and hidden initialisation. Take responses of the form

$$y_i = g_*(x_i) + \varepsilon_i.$$

Assume that the responses have a width-independent second-moment scale τ_y (e.g. $\mathbb{E}[y_i^2 | x_i] \leq \tau_y^2$, uniformly in x_i), and define

$$\tau_f^2 := \sup \{ \mathbb{E}_w [\sigma(w^\top x)^2] : \|x\| \leq L_\gamma \} < \infty.$$

Conditional on the data, the symmetric signs make each initial prediction centred with variance at most τ_f^2 . Summing the second moments and applying Markov's inequality gives, at any fixed confidence level,

$$\|F(W_0)\| = O(\tau_f \sqrt{n}), \quad \|y\| = O(\tau_y \sqrt{n}), \quad r_0 \leq C_r \sqrt{n},$$

where $C_r = O(\tau_f + \tau_y)$; the confidence-dependent constant does not depend on n , d , or m . Since σ is B_1 -Lipschitz,

$$\tau_f \leq |\sigma(0)| + B_1 L_\gamma.$$

Thus responses of constant-order magnitude give an initial Euclidean residual of size at most $O(\sqrt{n})$, at fixed confidence.

Combining these estimates with the preceding matrix Chernoff bound, for fixed $\delta \in (0, 1)$ the following estimates hold on the intersection of the data, kernel-concentration, and residual events:

$$\lambda_0 \geq (1 - \delta)\lambda_*, \quad \beta_{m,a} \leq \frac{C_\gamma}{\sqrt{m}}, \quad r_0 \leq C_r \sqrt{n}.$$

The initial-kernel concentration requires only

$$m \gtrsim \frac{B_1^2 L_\gamma^2}{\delta^2 \lambda_*} \log \frac{n}{\eta},$$

whereas the self-bounding condition is ensured by

$$\sqrt{\frac{m}{n}} > \frac{2C_\gamma C_r}{(1 - \delta)^2 \lambda_*^2}, \quad \text{or equivalently} \quad m > \frac{4C_\gamma^2 C_r^2}{(1 - \delta)^4 \lambda_*^4} n.$$

Thus, when the aspect ratio stays away from the singular threshold, the activation constants are order one, and the population kernel is well conditioned, the argument asks for width linear in the sample size, up to confidence factors. The appearance of the fourth power of λ_*^{-1} is a sign of the conservatism of controlling all kernel motion through its operator norm, and need not be sharp.

The same calculation displays the dependence on the output-weight and response scales. For general bounded output weights with a centred initial prediction, the estimates used in the width condition are

$$\lambda_0 \geq (1 - \delta)\nu_2(a)\lambda_*, \quad \beta_{m,a} \leq \frac{C_\gamma \nu_3(a)}{\sqrt{m}},$$

where the residual bound $r_0 \leq C_r \sqrt{n}$ must be recalibrated for the chosen output weights. Up to numerical and confidence factors, a sufficient width for the self-bounding condition is

$$m \gtrsim n \frac{C_\gamma^2 C_r^2 \nu_3(a)^2}{\lambda_*^4 \nu_2(a)^4}.$$

For fixed $\nu_2(a)$, the third moment $\nu_3(a)$ satisfies $\nu_3(a) \geq \nu_2(a)^{3/2}$ and is smallest when the magnitudes $|a_j|$ are constant. Symmetric Rademacher weights give a simple choice: their second moment is one, they minimise the drift coefficient at that second moment, and they keep the initial prediction centred.

Finally, the appearance of n becomes transparent if one writes the root-mean-square residual $\hat{r}_t := r_t/\sqrt{n}$. The comparison becomes

$$\dot{\hat{r}}_t \leq -\lambda_t \hat{r}_t, \quad \dot{\lambda}_t \geq -C_\gamma \sqrt{\frac{n}{m}} \hat{r}_t,$$

for symmetric Rademacher output weights. The pertinent small parameter is hence $\sqrt{n/m}$. Dividing the square loss by n likewise only rescales time and leaves the positivity condition unchanged.